One evening, two endings

A fictional but ordinary case: Thabo, 55, has type 2 diabetes and high blood pressure and is allergic to aspirin. At 18:40 he collapses at a taxi rank in Tembisa.

Time	Today, with paper folders	With MediChain
18:52 ambulance arrives	Thabo cannot speak. No one on scene knows his conditions, medicines or allergies.	The paramedic scans the QR on Thabo's phone lock screen, or taps his card: blood type, allergies, medicines and next of kin appear.
18:58 treatment starts	Standard chest-pain protocols often include aspirin, so a responder must stop and ask. Nobody can answer.	The aspirin allergy is the first thing on screen. The grant lasts 15 minutes and covers Thabo only.
19:30 hospital handover	His folder is at his clinic, 20 km away. Staff phone family members.	The receiving doctor opens the same record: notes, recent results and current medicines.
Next morning	Thabo has no way to know who read his information.	Thabo's access history lists the paramedic and the doctor, with times and reasons.

This is an illustrative scenario, not a measured outcome. MediChain makes no claim about mortality or error rates; those will be measured in a pilot.

The evidence behind the problem

- **Limpopo, 40 public hospitals:** missing records "negatively affected timely and effective healthcare service delivery"; staff were "unable to assist patients or treat them immediately" [1].
- **National case law, 2024:** poor record-keeping undermines patients' own claims, because facilities fail their legal obligations [2].
- **Khayelitsha Hospital emergency centre:** months of folders stayed un-scanned and unavailable electronically [3] (one hospital, 2012–2013).
- **Policy:** the National Digital Health Strategy calls for a complete electronic record and a common identifier [4].

Who gains what

Who	Before	After
Patients	History scattered across folders; no view of who reads it	Carry their own summary; approve and withdraw access; read their notes, results and messages
Paramedics	Treat blind or wait for a phone call	Allergies and medicines on scene, on an approved device
Doctors and nurses	Re-take histories; handovers on paper	One shared record; notifications only about their own patients
Pharmacists and laboratories	Prescriptions and requests that cannot be traced	Prescriptions and specimens tied to the patient and the prescriber
Facilities	Disputes argued from missing paper	A signed, write-once record of every access

Built for South African reality

- **Load-shedding and weak signal:** the QR carries a printed summary that still shows offline, marked unverified.
- **Any phone:** web apps that install from the browser, with no app-store download or data-heavy update.
- **Local hosting:** data stays in South Africa (AWS Cape Town).
- **Six languages prepared:** English, isiZulu, Kiswahili, Hausa, Amharic and French interface bundles exist; English is complete.

How a pilot will prove it

Measure	How
Seconds to critical information	From first contact to allergies on screen, against the current route
Handover completeness	Share of handovers with allergies, medicines and contacts recorded
Patient understanding	Can patients say who opened their record, and why?
Adoption	Active staff per shift; workflows finished without paper

Sources

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3. Ohuabunwa, E. U., Sun, J., Jubanyik, K. J. & Wallis, L. A. (2016). Electronic medical records in low to middle income countries: the case of Khayelitsha Hospital, South Africa. *African Journal of Emergency Medicine*, 6(1), 38–43. https://doi.org/10.1016/j.afjem.2015.06.003
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